Afaf Binta Shamsuddin

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Education

Bachelor of Science in Computer Science Jan. 2022 – Jan. 2026
BRAC University *Dhaka, Bangladesh*
- Relevant Coursework: DBMS, System Design, Networking, French, German

Higher School Certificate (HSC), Science 2018 – 2020
Shaheed Bir Uttam Lt. Anwar Girls' College *Dhaka, Bangladesh*
- GPA: **5.00**/5.00

Secondary School Certificate (SSC), Science 2016 – 2018
Shaheed Bir Uttam Lt. Anwar Girls' College *Dhaka, Bangladesh*
- GPA: **5.00**/5.00
- Achived Best Student Performance Award 2018 based on merit.

Experience

Trainee Software Engineer Apr. 2026 – Present
National Polymer Group *Dhaka, Bangladesh*
- Maintained in-house Sales, Dealer, and Office applications used by **1,200+ users** across the organization.
- Redesigned the Sales app using newer, efficient packages, cutting size from **35 MB to 17 MB** and boosting speed up to **50%**.
- Shipped **6+ new Sales app features**, including conveyance bill submission, an analytics dashboard, and pause-order delivery.
- Migrated the Office app from React Native to Kotlin, improving performance and reducing app size from **19 MB to 5 MB**.

Projects

AdAstra (Astronomical Image Classification & RAG Platform) *(Team of 4)* 2026
PyTorch, EfficientNet-B0, CLIP, FAISS, Gemma, Vercel, MongoDB Atlas Live
- Built a two-stage CNN recognition pipeline (EfficientNet-B0) classifying astronomical images into **5 object types** and galaxy morphology, reaching **83.7% test accuracy** on SpaceNet and **86.9%** on Galaxy Zoo across **70,000+ training images**, with a CLIP-based gate rejecting non-astronomical inputs and a confidence rule flagging uncertain predictions.
- Designed a retrieval-augmented explanation engine: a FAISS index over **6,222 chunks** from 113 curated Wikipedia articles feeds the Gemma language model to generate citation-grounded, multi-level (beginner/intermediate/advanced) explanations, with post-generation citation verification to strip unsupported references.
- Shipped a full-stack web platform (Vercel backend API, MongoDB Atlas) with authenticated accounts, saved analysis history, a source-cited astronomy Q&A assistant, and a sky explorer / Moon-phase page, serving concurrent requests via a load-balanced architecture.

Extracurricular Activities

Senior Executive 2023 – 2025
BRAC University Art & Photography Society *Dhaka, Bangladesh*

General Secretary 2016 – 2017
SAGC Science Club *Dhaka, Bangladesh*

Technical Skills

Languages: Python, Java, Kotlin, JavaScript, SQL, Bash | **Databases:** Oracle, PostgreSQL, MySQL
Frameworks: React, Laravel, Flask | **Systems:** Linux (CLI, Bash scripting)
Tools: Git, GitHub Actions, Docker, Postman, Jira, Swagger | **Testing:** Selenium, k6, Unit Testing

Achievements and Certifications

DRMC-Savlon National Science Carnival X (2017) — **1st Place**, Project Display Competition, secondary level, organized by Dhaka Residential Model College.